

[CSED211] Introduction to Computer Software Systems

Lab 6: Shell Lab

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Today's Agenda

- Shell
- Shell Lab
- Quiz

What is shell?

- An interactive command-line interpreter that runs programs on behalf of user
- **Command line**: a sequence of ASCII text words
- **Common examples**: Bash(Bourne-again shell)
 - Linux default
- Most applications in linux (command line) are run through shell

Basic functions of shell

- **jobs**
 - List the running and stopped background job
- **bg <job>**
 - Change a stopped background job to running background job
- **fg <job>**
 - Change a stopped or running background job to running in the foreground
- **ctrl+c**
 - Deliver *SIGINT* signal to each process in the foreground job
- **ctrl+z**
 - Deliver *SIGSTP* signal to each process in the foreground job

Basic functions of shell

- Example

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1] Stopped vi 111
[2]- Stopped vi 222
[3]+ Stopped vi 333
```

Job id, spec

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1] Stopped vi 111
[2]- Stopped vi 222
[3]+ Stopped vi 333
```

status

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1] Stopped vi 111
[2]- Stopped vi 222
[3]+ Stopped vi 333
```

Job name

Basic functions of shell

- Example

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1]  Stopped          vi 111
[2]-  Stopped          vi 222
[3]+  Stopped          vi 333
```

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ fg %1
vi 111
```

↳ Change a stopped background job to running in the foreground

Basic functions of shell

- Example
 - If use ctrl+z(**^Z**) foreground task will stop
 - If you want to run task again, use **fg %1** or **bg %1**

```
[1]+  Stopped          vi 111
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1]+  Stopped          vi 111
[2]   Stopped          vi 222
[3]-  Stopped          vi 333
```

Basic functions of shell

- Example

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ python count_10.py
start
```

```
^Z
[1]+  Stopped                  python count_10.py
```

Task stop

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1]+  Stopped                  python count_10.py
```

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ bg %1
```

Task run again in background

```
[1]+  python count_10.py &
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
[1]+  Running                  python count_10.py &
```

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ python count_10.py
start
```

```
^C
Traceback (most recent call last):
  File "count_10.py", line 5, in <module>
    time.sleep(1)
```

Task kill

```
KeyboardInterrupt
```

```
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$ jobs
hcle@Hcle:/mnt/c/Users/Hcle/Desktop$
```


Shell Lab

- Write a simple shell ('tsh')
- Only need to write **seven** specified functions
- **Helper functions** already provided in source file

tsh basic functionality

- Run a command or application on shell
- e.g., `tsh>/bin/ls -l -h`
 - Shell forks child process
 - Executes “`/bin/ls`” with arguments “`-l`” and “`-h`”
 - `argv[0] = “/bin/ls”, argv[1] = “-l”, argv[2]=“-h”`
- **tsh** manages running application as **child process**

tsh basic functionality

- Foreground job management
 - Runs application in foreground and waits for its ending
- e.g., **tsh>/bin/ls -l -h**
 - Shell executes “/bin/ls” with “-l -h”
 - Shell Wait for it to finish before other application runs
- Every application run is foreground by default

tsh basic functionality

- Background job management
 - Runs application in background
 - Many **simultaneous background** jobs possible
- “&” added to end of command/application name
- e.g., **tsh> ./myprogram &**

tsh basic functionality

- Background/Foreground management
 - Changes job status (**bg** to **fg** / **fg** to **bg**)
- e.g., **tsh>/fg <job_id>**
 - Makes a background job runs as foreground
- e.g., **tsh>/bg <job_id>**
 - Makes a foreground job run as background

Work in this assignment

- Many **helper functions** coded for you
 - Parseline
 - addjob, deletejob, clearjob
 - Fqpid
 - getjobpid, getjobid
- 4 programs use by **tsh**
 - myint
 - myspin
 - mysplit
 - mystop

Work in this assignment

- **eval**
 - Main routine that parses and interprets the command line
- **builtin_cmd**
 - Recognizes and interprets the built-in commands
 - e.g., **quit**, **fg**, **bg**, and **jobs**
- **do_bgfg**
 - Implements the **bg** and **fg** built-in commands
- **waitfg**
 - Waits for a foreground job to complete

Work in this assignment

- `sigchld_handler`
 - Catches *SIGCHLD* signals
- `sigint_handler`
 - Catches *SIGINT*(ctrl+c) signals
- `sigstp_handler`
 - Catches *SIGSTP*(ctrl+z) signals

Work in this assignment

- `sigchld_handler`
 - Catches *SIGCHLD* signals

• *Reuse* already coded functions in “*tsh.c*” to write specified functions

How to evaluate your code

- Use the provided **reference solutions** `tshref` and 16 traces
- Test each `tsh` feature with a trace and provided `sdriver.pl`
- Your `tsh` output must match `tshref.out`
 - The PIDs can (and will) be different.
- **Evaluate order**
 - Assuming you are evaluating trace01, you need to run two commands as shown below
 - `./sdriver.pl -t trace01.txt -s ./tsh -a "-p" or make test01`
 - `./sdriver.pl -t trace01.txt -s ./tshref -a "-p" or make rtest01`
 - If the result of ① and ② match, then your code is correct

Tips

- You should use `kill()` function well
- `fork()` return 0 in child process, but return pid of child process in parent process
 - You can distinguish child or parent by checking the return value
 - `fork()` also copies signal mask from parent to child
- You can use `waitpid()` function
 - You can wait child process and check child's status

Evaluation

- Deadline: 12/8 (Mon) 23:59 (midnight)
- Score:
 - 10 pts Quiz.
 - 40 pts Correctness: 16 trace files at 2.5 points each.
 - 50 pts Report.

Cheating Policy

- You **can refer**
 - Shell lab writeup, lab slides, lecture slides
- You **should not refer**
 - ChatGPT
 - Code and report from a senior who has already taken this course
 - Blogs or github repository that contain solution codes
 - Every other references that violate POSTECHIAN's Honor

Quiz

- Go to the PLMS, and start the quiz!